Age and Medical Conditions

Gender-Stratified Prevalence

Raw Data - No Fitted Lines

HPP 10K Dataset

Ages 40-70

Female: 12,313 | Male: 11,152

Blue = Male | Red = Female

EXECUTIVE SUMMARY: GENDER-STRATIFIED PREVALENCE

This report shows RAW PREVALENCE DATA ONLY - no statistical models or fitted lines. Each point represents the actual percentage of people with each condition in that age bin.

SAMPLE SIZE

- Female: 12,313 subjects (52%)
- Male: 11,152 subjects (48%)
- Age range: 40-70 years
- Age bins: 40-45, 45-50, 50-55, 55-60, 60-65, 65-70

WHAT YOU SEE IN THE PLOTS

- Blue circles = Male prevalence at each age bin
- Red squares = Female prevalence at each age bin
- Error bars = 95% confidence intervals (Wilson score)
- NO fitted lines or statistical models

TOP CONDITIONS - MALES (Largest Increase 40→70)

- Hyperlipidemia: 13.1% → 38.9%
- Hypertension: 5.1% → 28.9%
- Prediabetes: 3.9% → 15.1%
- Hearing loss: 3.4% → 9.0%
- Ischemic Heart Disease: 0.7% → 5.5%

TOP CONDITIONS - FEMALES (Largest Increase 40→70)

- Hyperlipidemia: 7.1% → 34.9%
- Hypertension: 2.7% → 21.8%
- Prediabetes: 3.7% → 14.2%
- Back Pain: 12.0% → 20.7%
- Osteoarthritis: 0.8% → 8.0%

KEY OBSERVATIONS

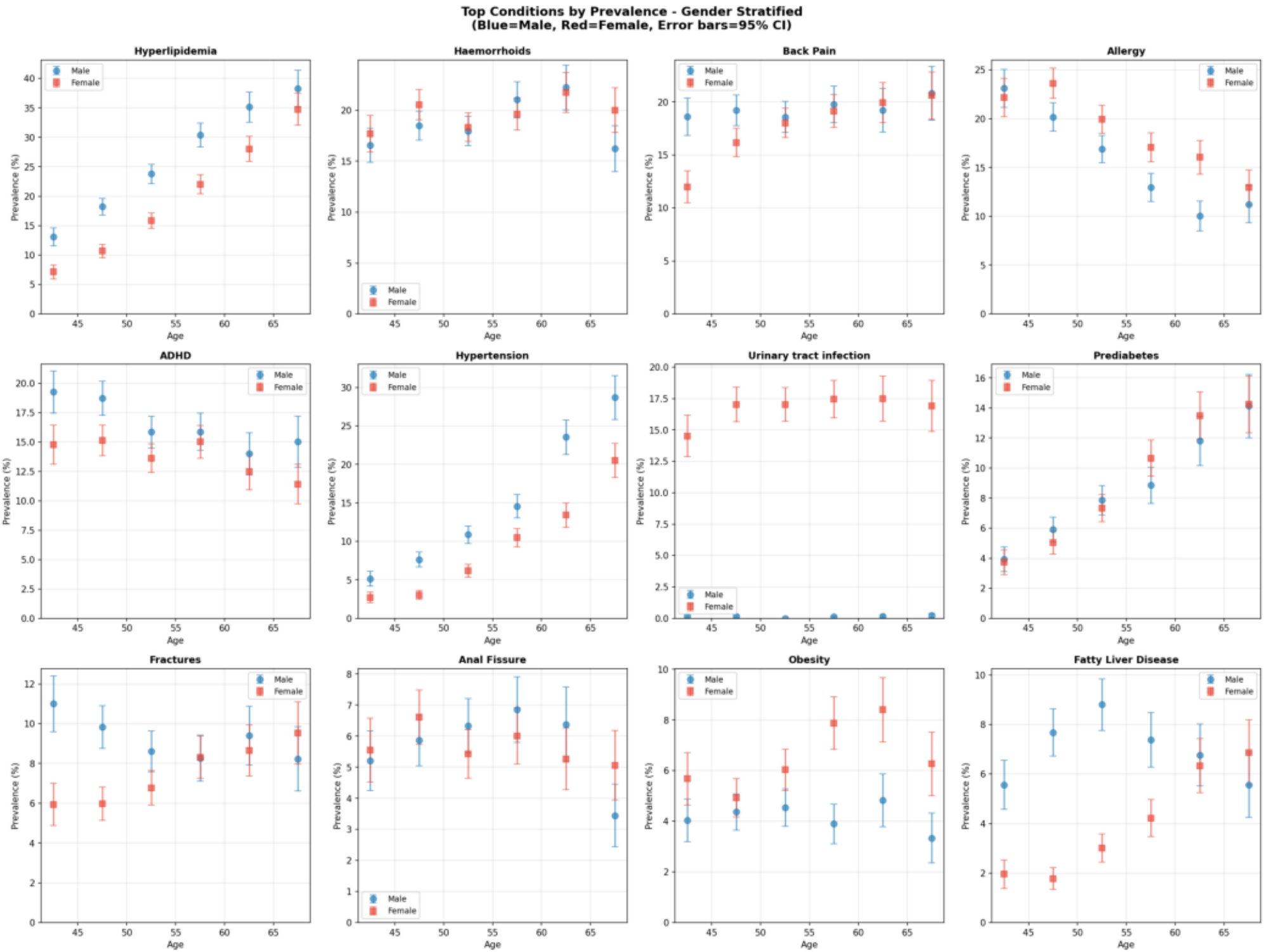
1. CARDIOVASCULAR CONDITIONS
   Hypertension, Hyperlipidemia show clear increases with age in BOTH genders, with males often having higher prevalence.

2. METABOLIC CONDITIONS
   Diabetes, Prediabetes increase with age in both genders.

3. MENTAL HEALTH
   Depression, Anxiety, ADHD show DECREASES with age - likely cohort effects (younger generations diagnosed more often).

4. SEX-SPECIFIC CONDITIONS
   Some conditions only appear in one gender (e.g., PCOS).

Top Conditions by Prevalence

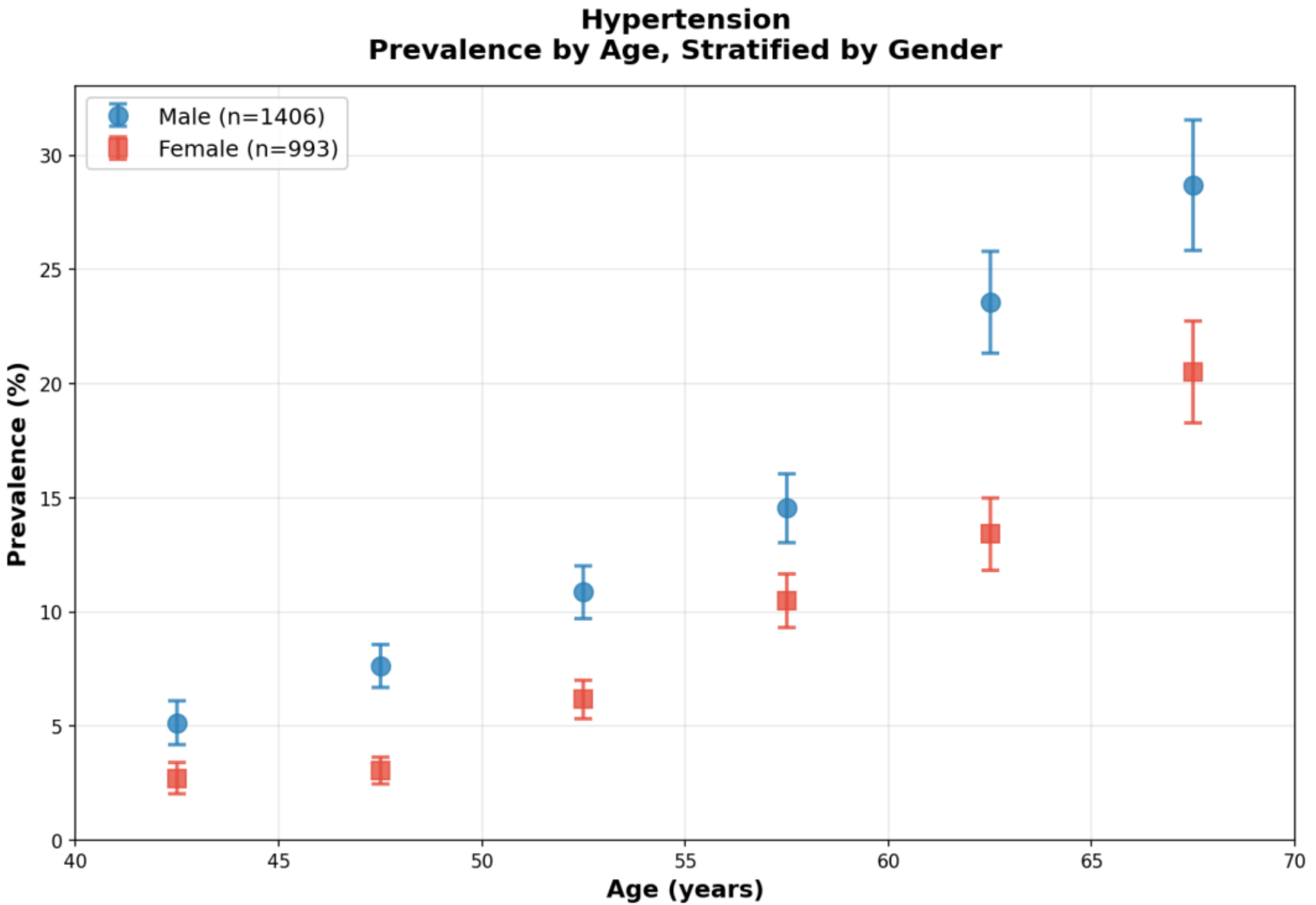


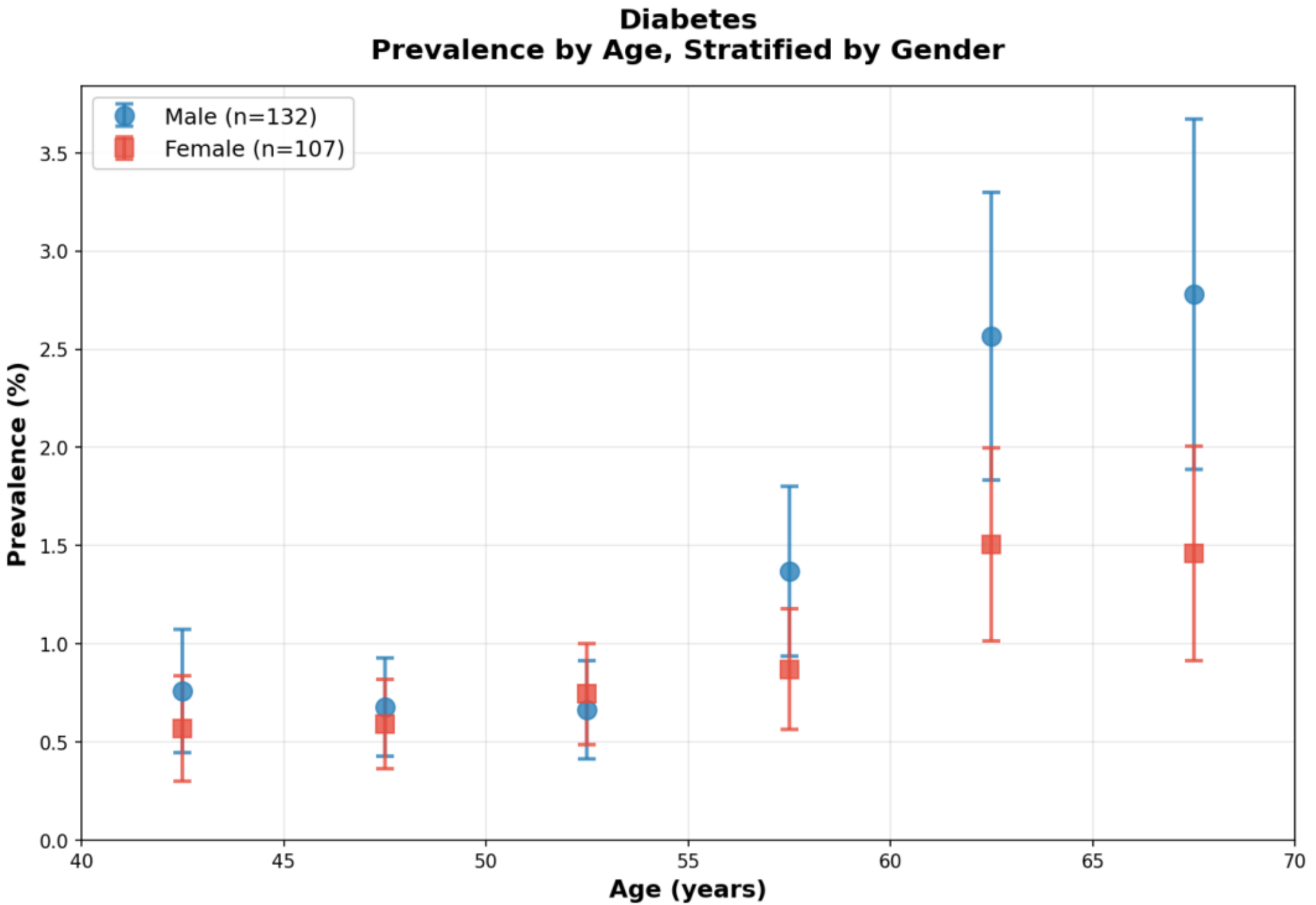
Top 15 Conditions - MALES (Largest Increase with Age)

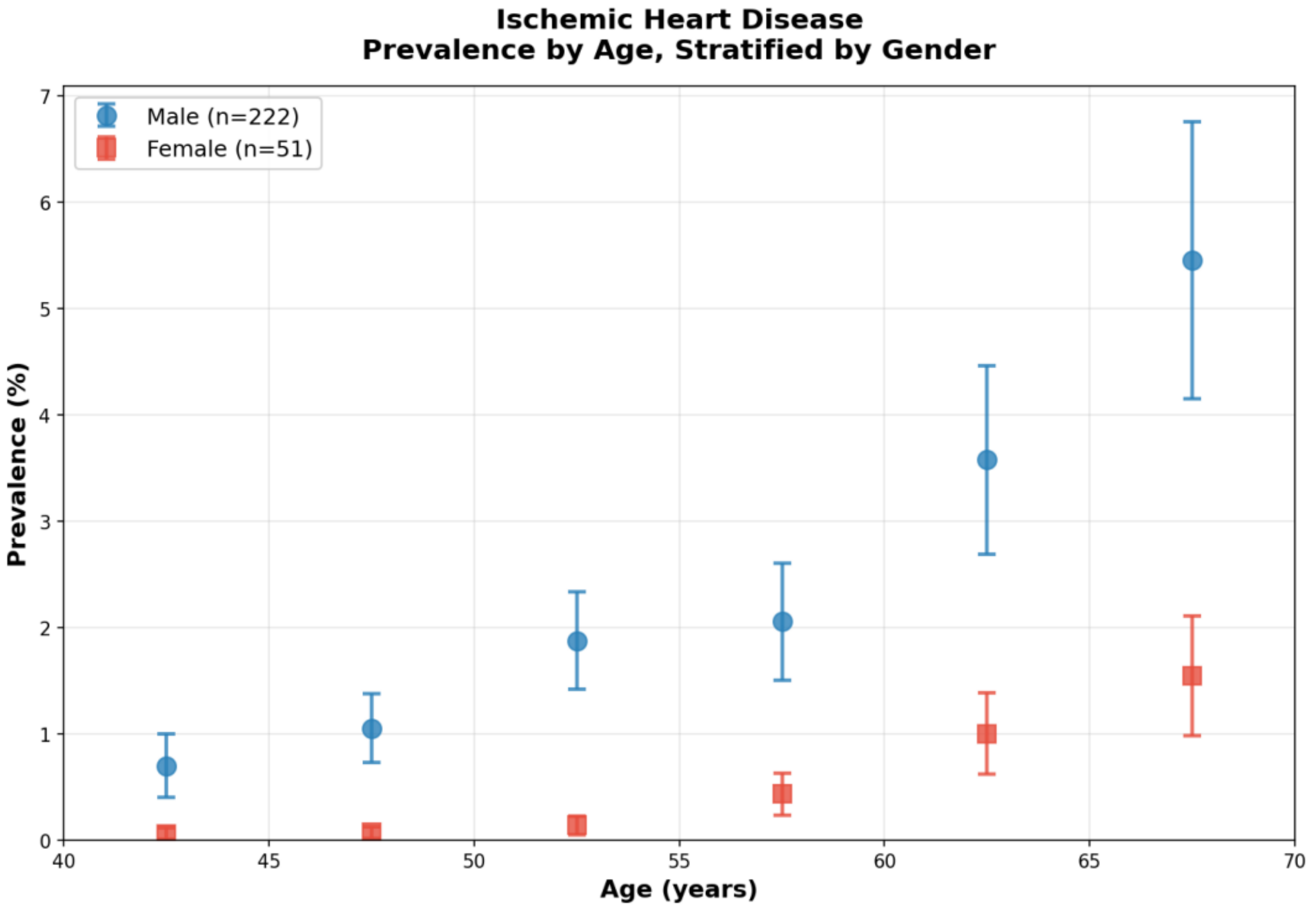
Condition	N	Prev@40 (%)	Prev@70 (%)	Change (pp)
Hyperlipidemia	2751	13.1	38.9	25.8
Hypertension	1451	5.1	28.9	23.8
Prediabetes	903	3.9	15.1	11.2
Hearing loss	457	3.4	9.0	5.6
Ischemic Heart Disease	231	0.7	5.5	4.8
Osteoarthritis	216	0.9	4.5	3.6
Urinary Tract Stones	510	2.9	6.1	3.2
Sleep Apnea	490	3.5	6.6	3.1
Gallstone Disease	256	1.6	4.1	2.5
Erectile Dysfunction	95	0.1	2.6	2.5
Back Pain	2142	18.6	20.7	2.1
Diabetes	137	0.8	2.9	2.1
Retinal detachment	76	0.4	2.4	2.0
Gout	197	1.0	3.0	2.0
Glaucoma	77	0.1	1.7	1.6

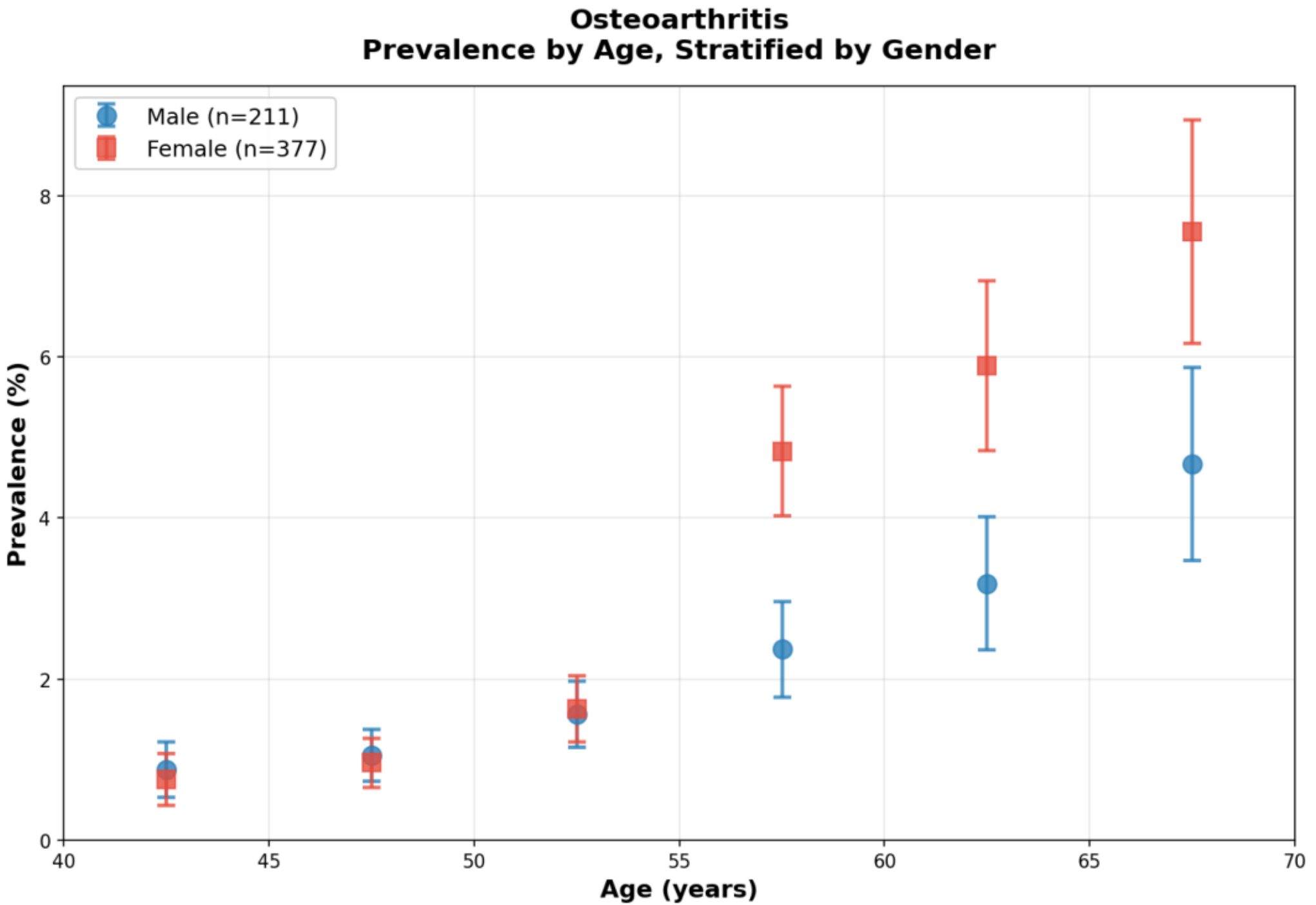
Top 15 Conditions - FEMALES (Largest Increase with Age)

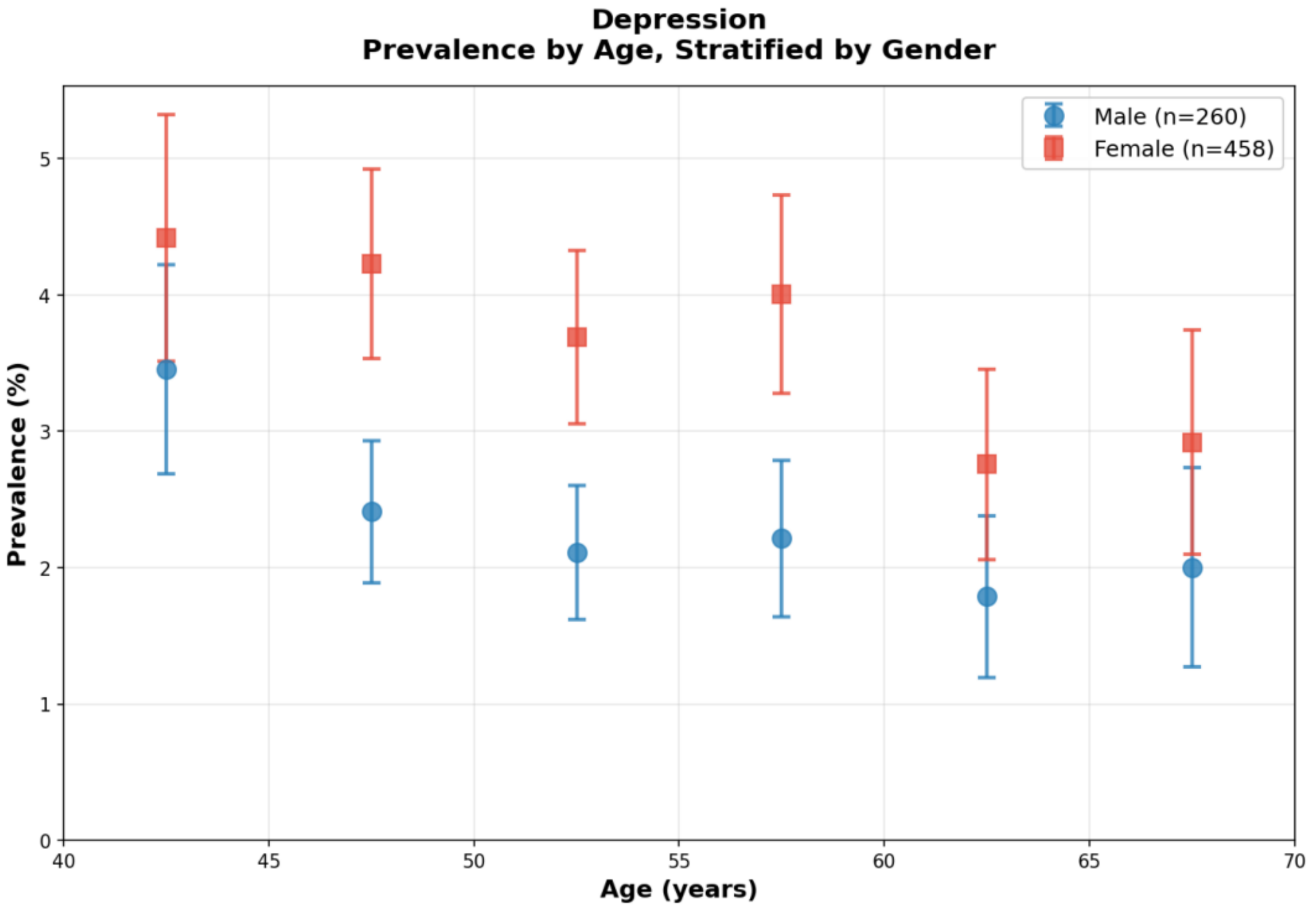
Condition	N	Prev@40 (%)	Prev@70 (%)	Change (pp)
Hyperlipidemia	2261	7.1	34.9	27.8
Hypertension	1042	2.7	21.8	19.1
Prediabetes	1049	3.7	14.2	10.4
Back Pain	2165	12.0	20.7	8.7
Osteoarthritis	395	0.8	8.0	7.3
Fatty Liver Disease	456	2.0	7.1	5.2
Hearing loss	365	1.5	5.5	4.0
Fractures	902	5.9	9.6	3.7
Gallstone Disease	629	4.0	7.4	3.4
Urinary tract infection	2069	14.5	16.7	2.1
Sleep Apnea	123	0.3	2.3	2.1
Insomnia	211	0.7	2.6	2.0
Peptic Ulcer Disease	205	0.9	2.8	1.9
Retinal detachment	124	0.4	2.3	1.8
Glaucoma	84	0.3	2.0	1.7

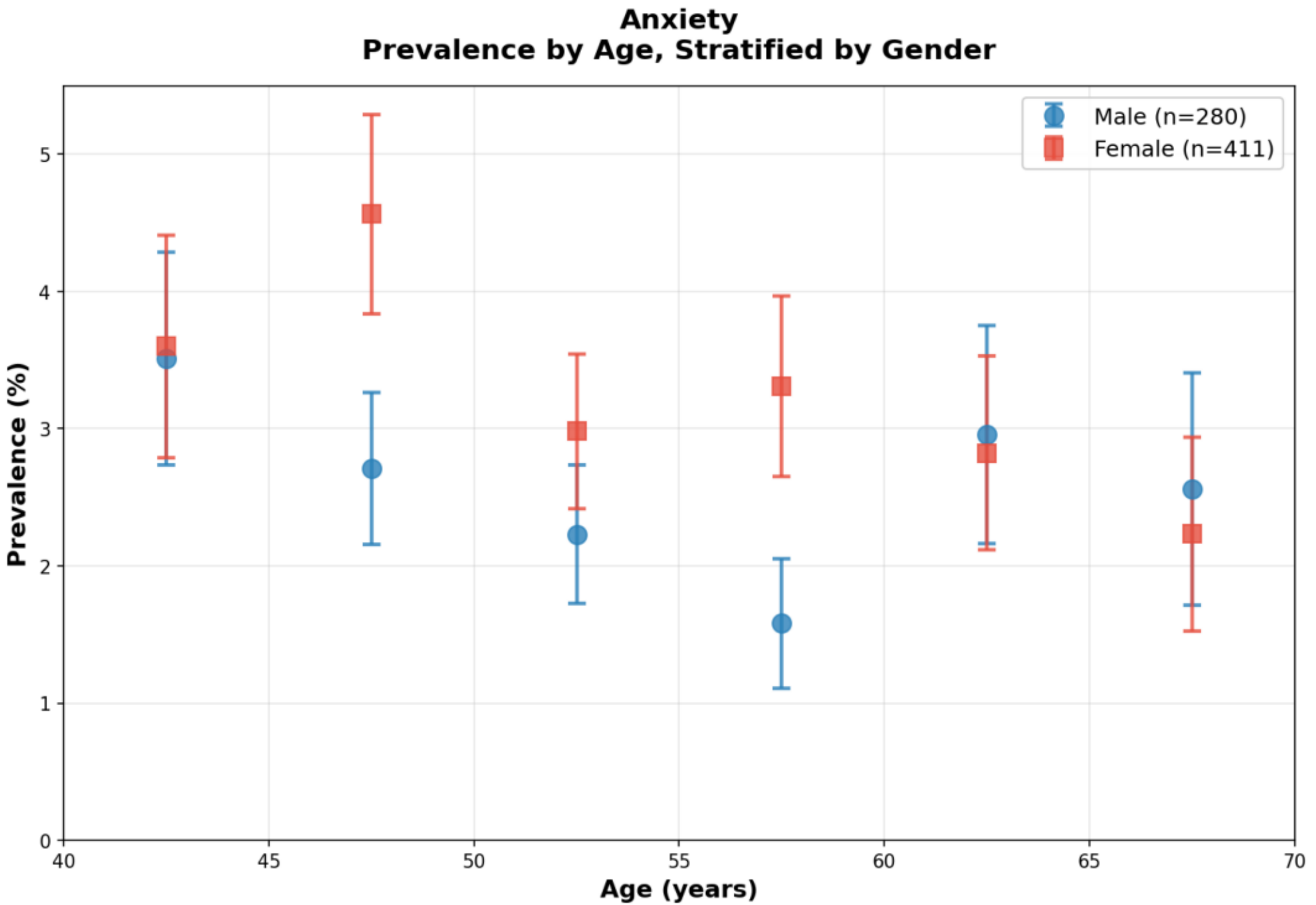

Hypertension
Prevalence by Age, Stratified by Gender
Male (n=1406)
Female (n=993)
Prevalence (%)
Age (years)
0
5
10
15
20
25
30
40
45
50
55
60
65
70

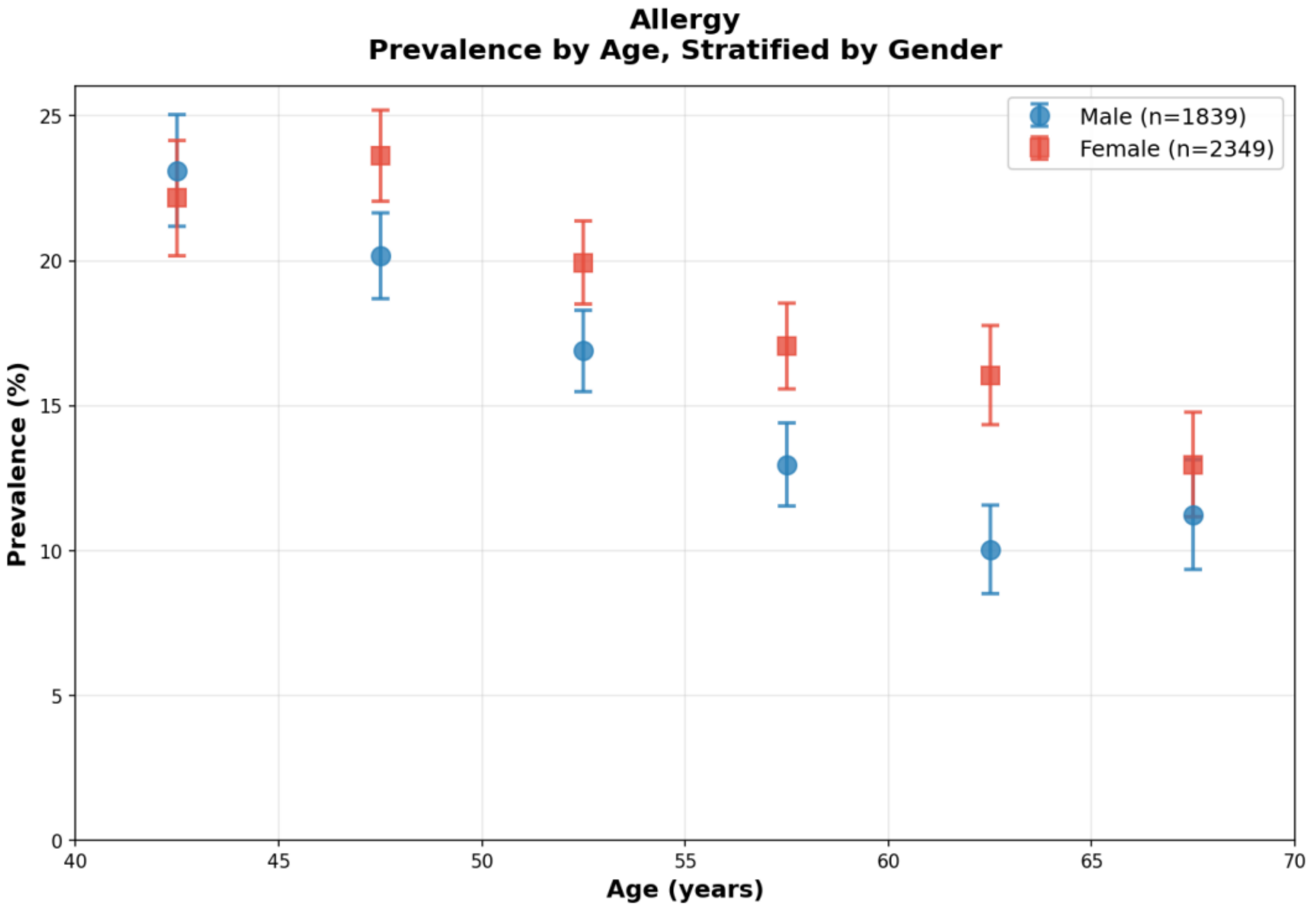

Diabetes
Prevalence by Age, Stratified by Gender
Male (n=132)
Female (n=107)
Prevalence (%)
Age (years)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
40
45
50
55
60
65
70

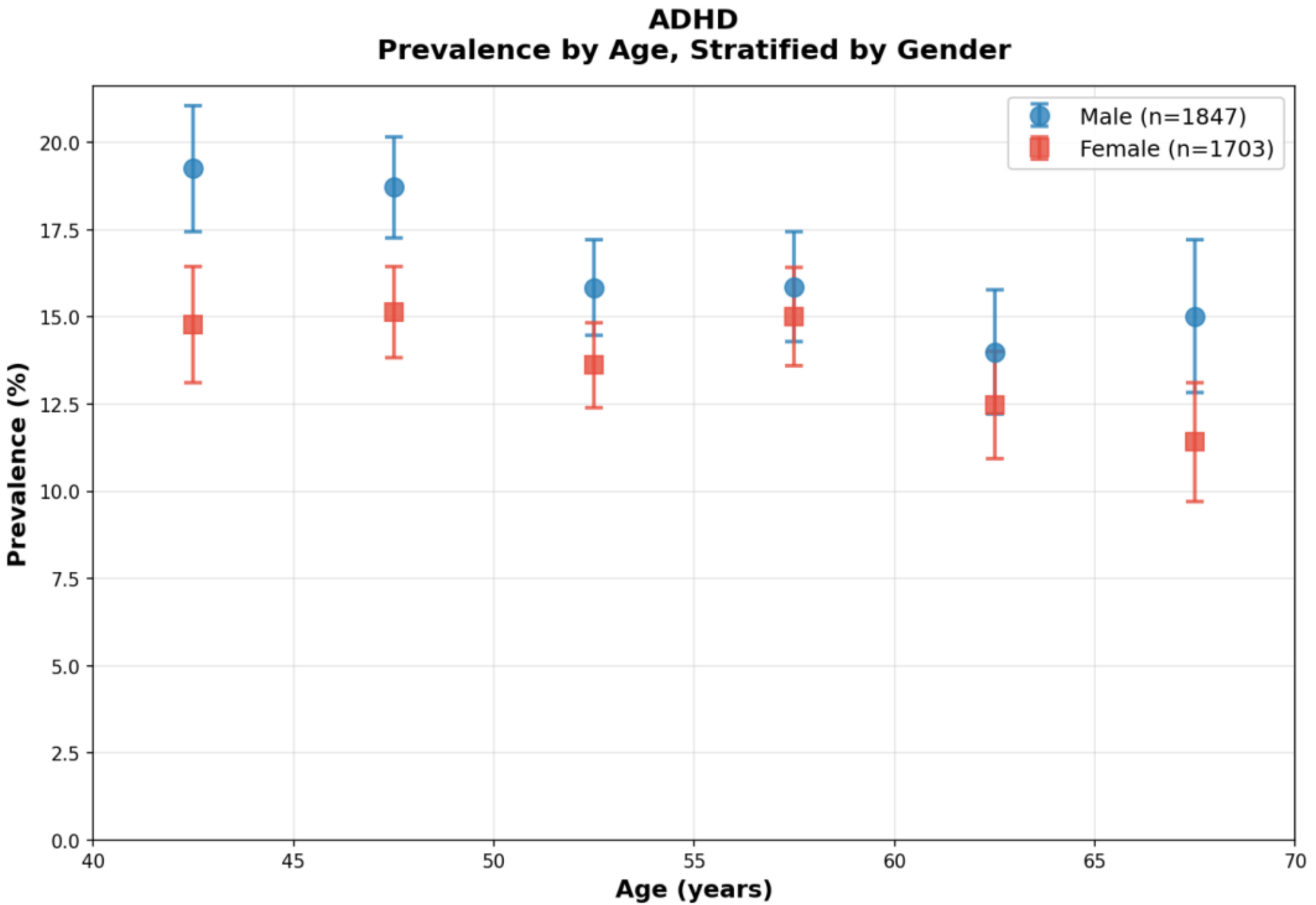

Ischemic Heart Disease
Prevalence by Age, Stratified by Gender
Male (n=222)
Female (n=51)
Prevalence (%)
Age (years)
0
1
2
3
4
5
6
7
40
45
50
55
60
65
70

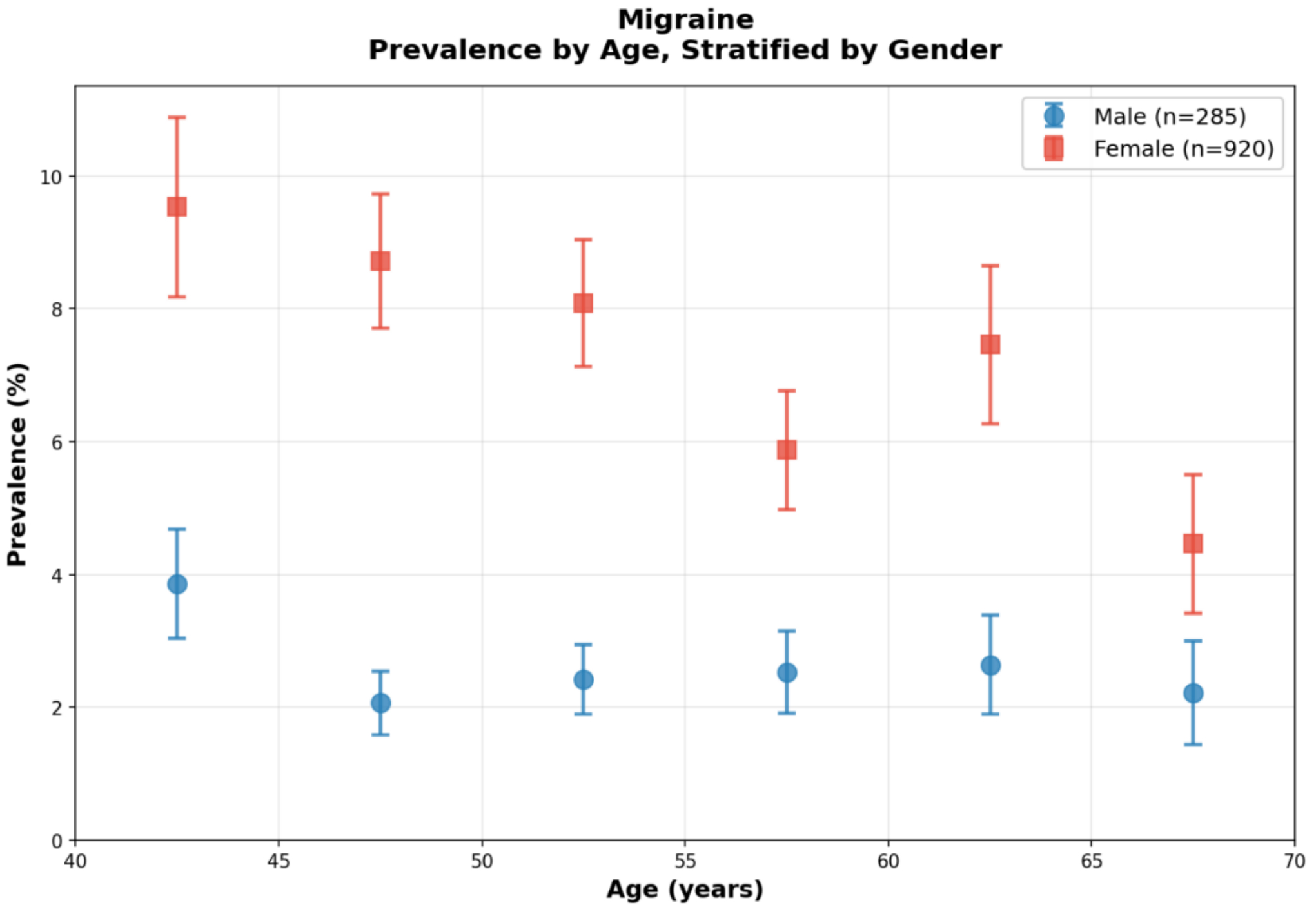

Osteoarthritis
Prevalence by Age, Stratified by Gender
Male (n=211)
Female (n=377)
Prevalence (%)
0
2
4
6
8
40
45
50
55
60
65
70
Age (years)

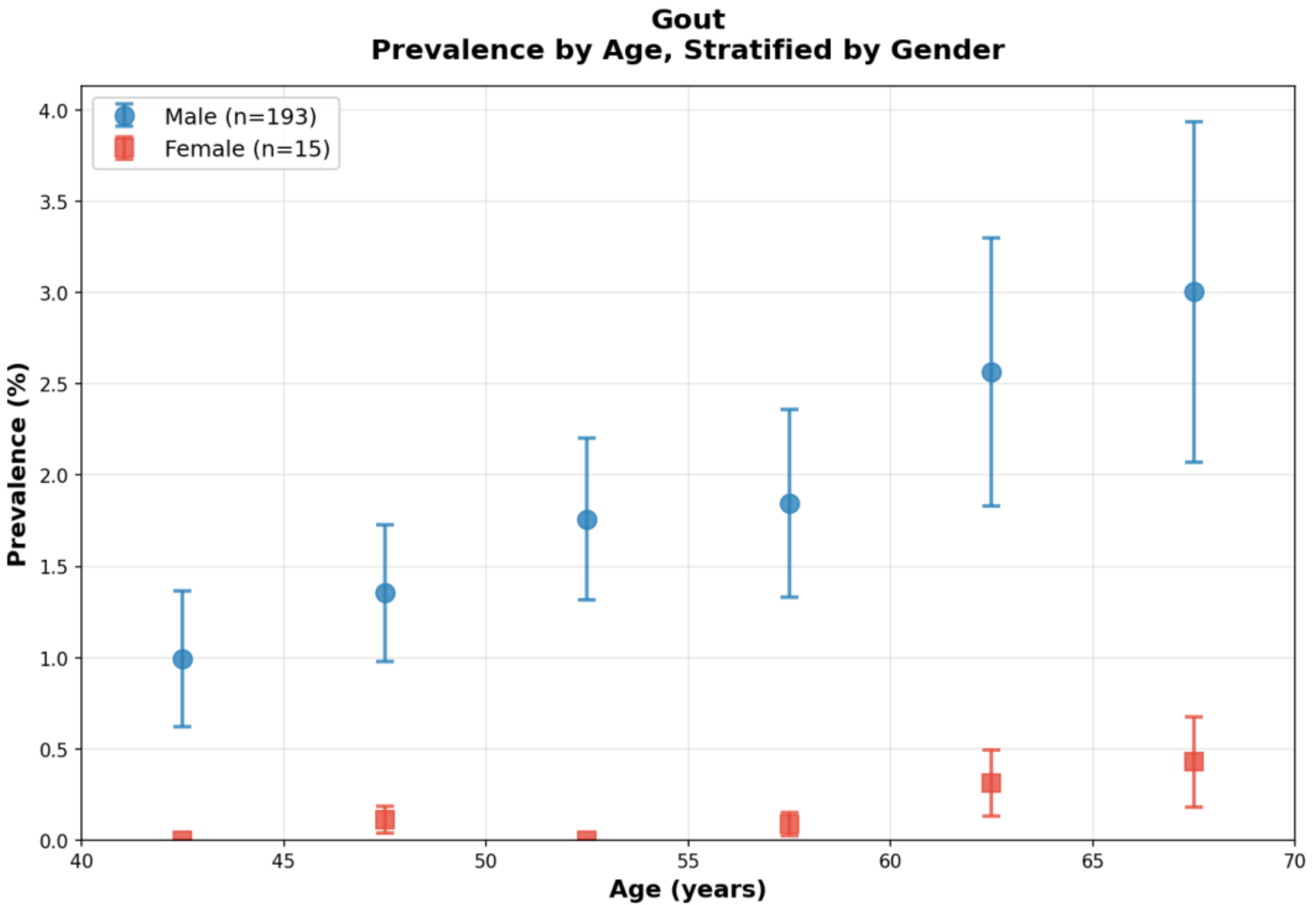

Depression
Prevalence by Age, Stratified by Gender
Male (n=260)
Female (n=458)
Prevalence (%)
Age (years)
0
1
2
3
4
5
40
45
50
55
60
65
70

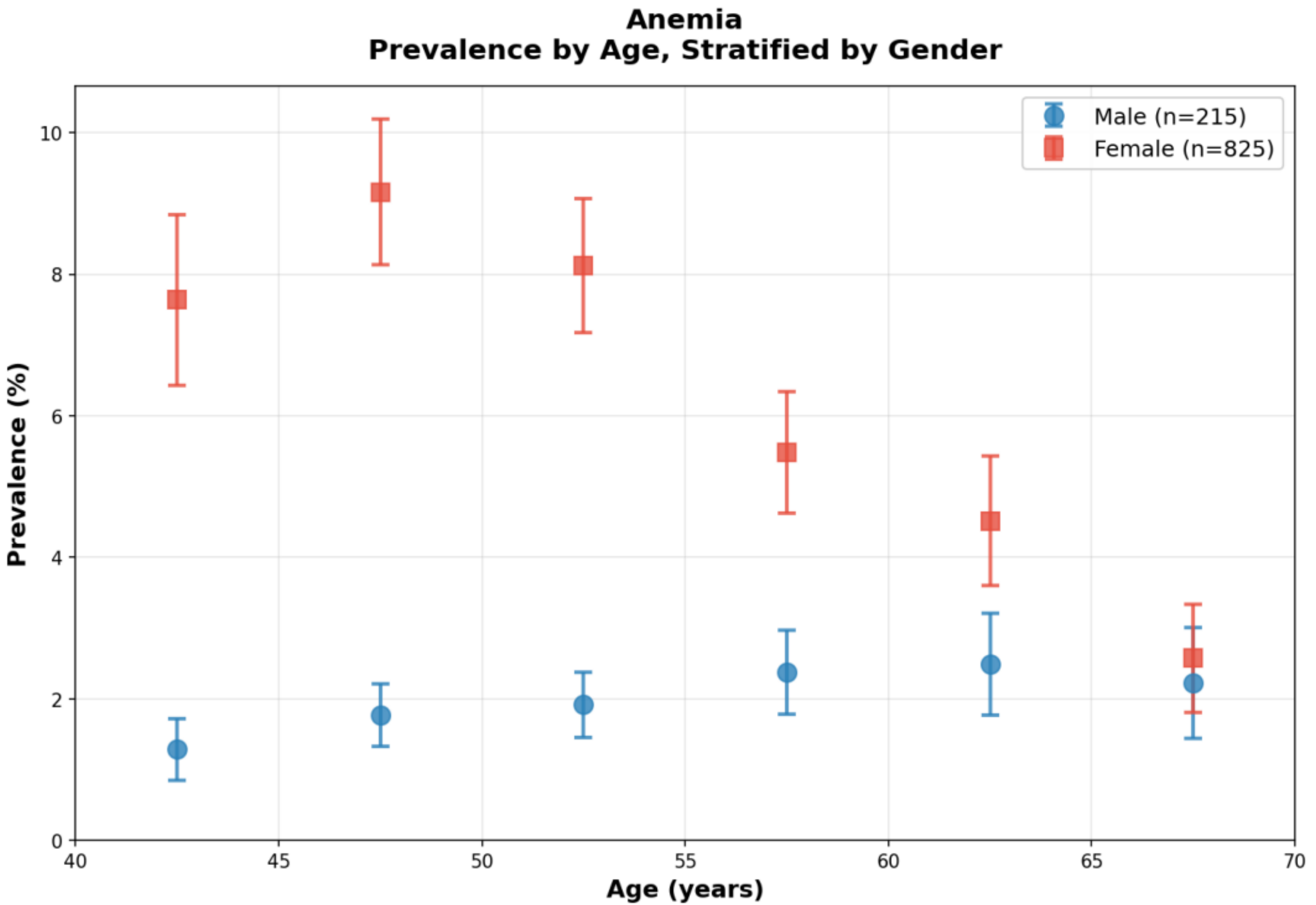

Anxiety
Prevalence by Age, Stratified by Gender
Male (n=280)
Female (n=411)
Prevalence (%)
Age (years)
0
1
2
3
4
5
40
45
50
55
60
65
70

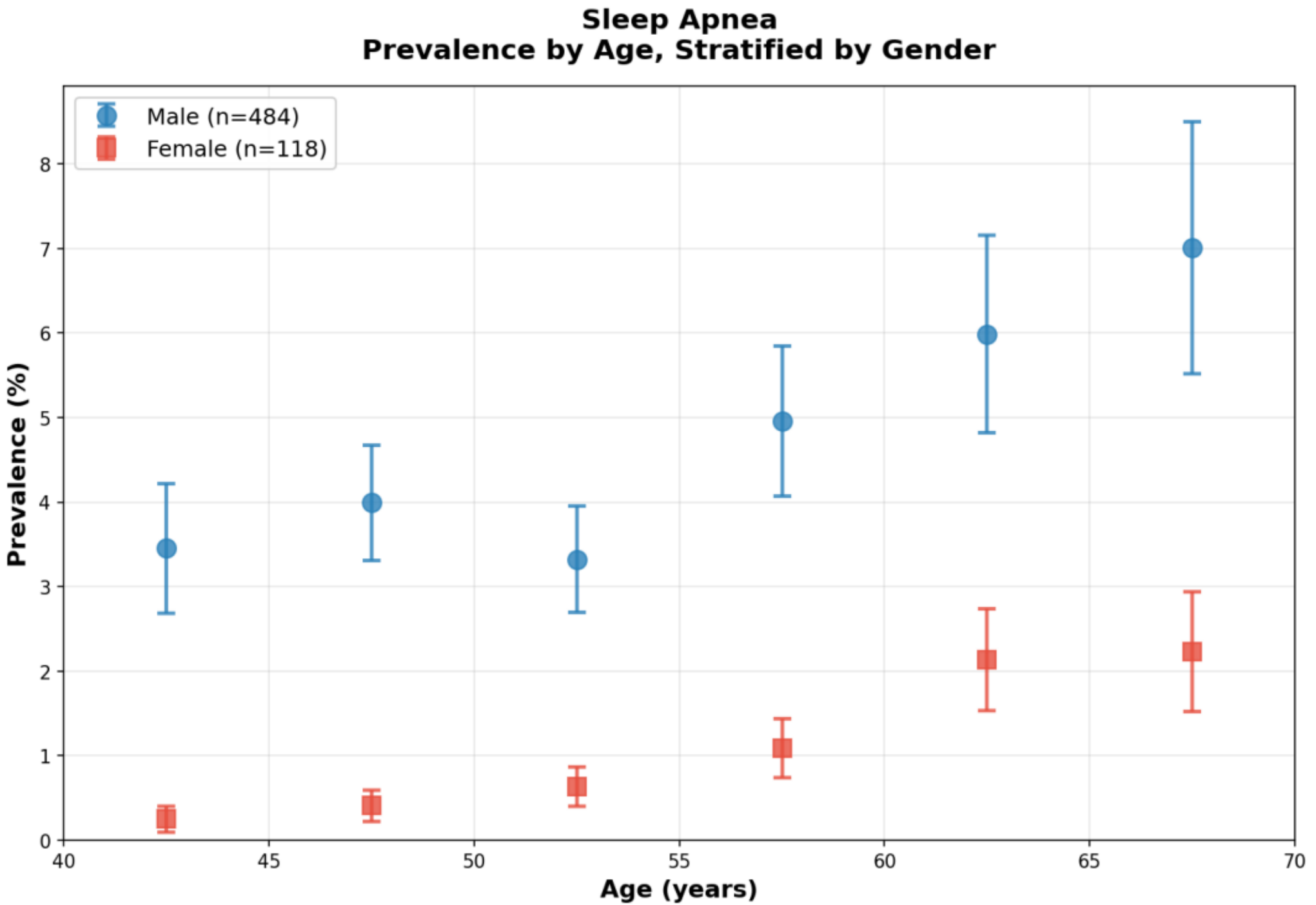

Allergy
Prevalence by Age, Stratified by Gender
Male (n=1839)
Female (n=2349)
Prevalence (%)
Age (years)
0
5
10
15
20
25
40
45
50
55
60
65
70

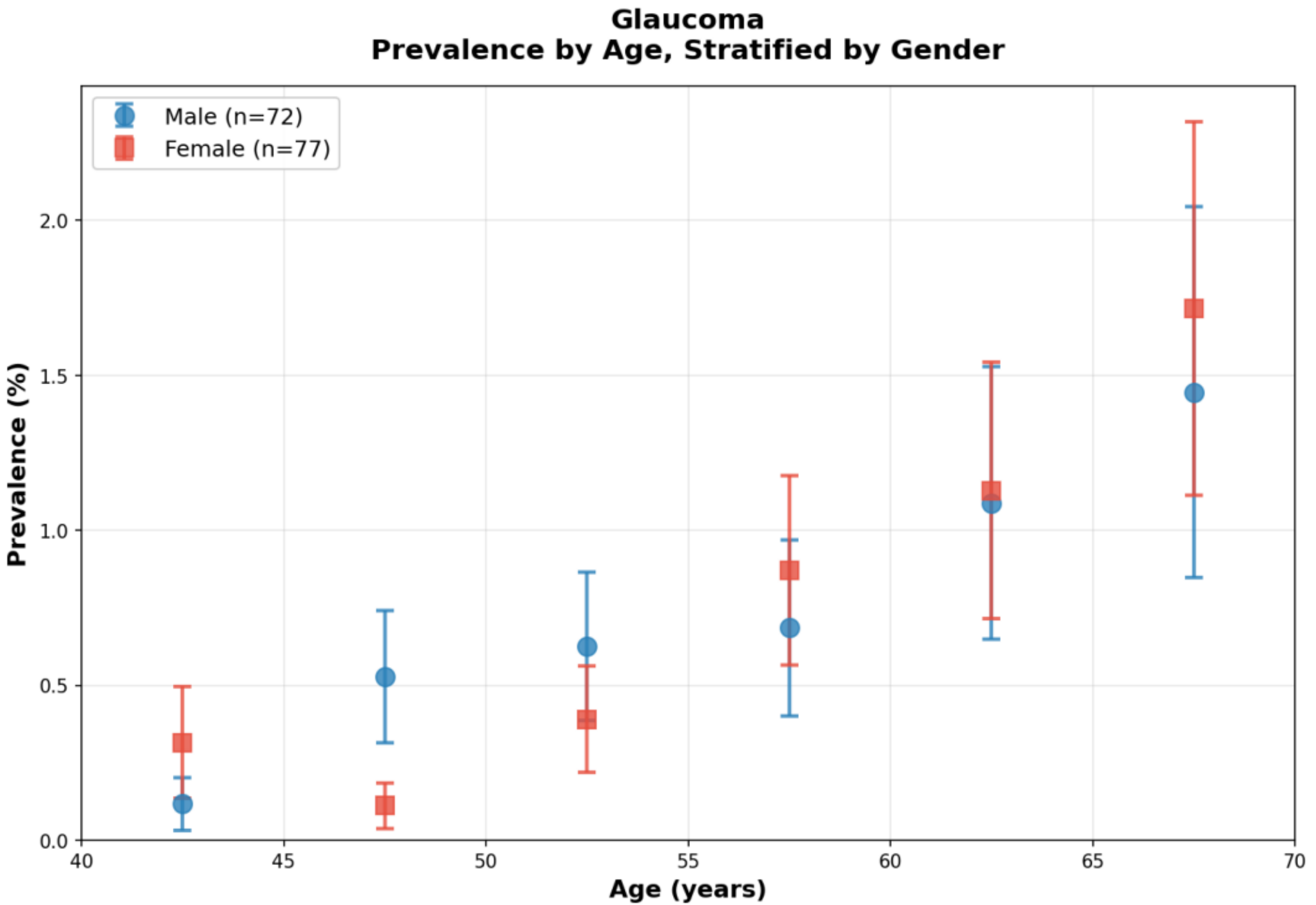

ADHD
Prevalence by Age, Stratified by Gender
Male (n=1847)
Female (n=1703)
Prevalence (%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
40
45
50
55
60
65
70
Age (years)

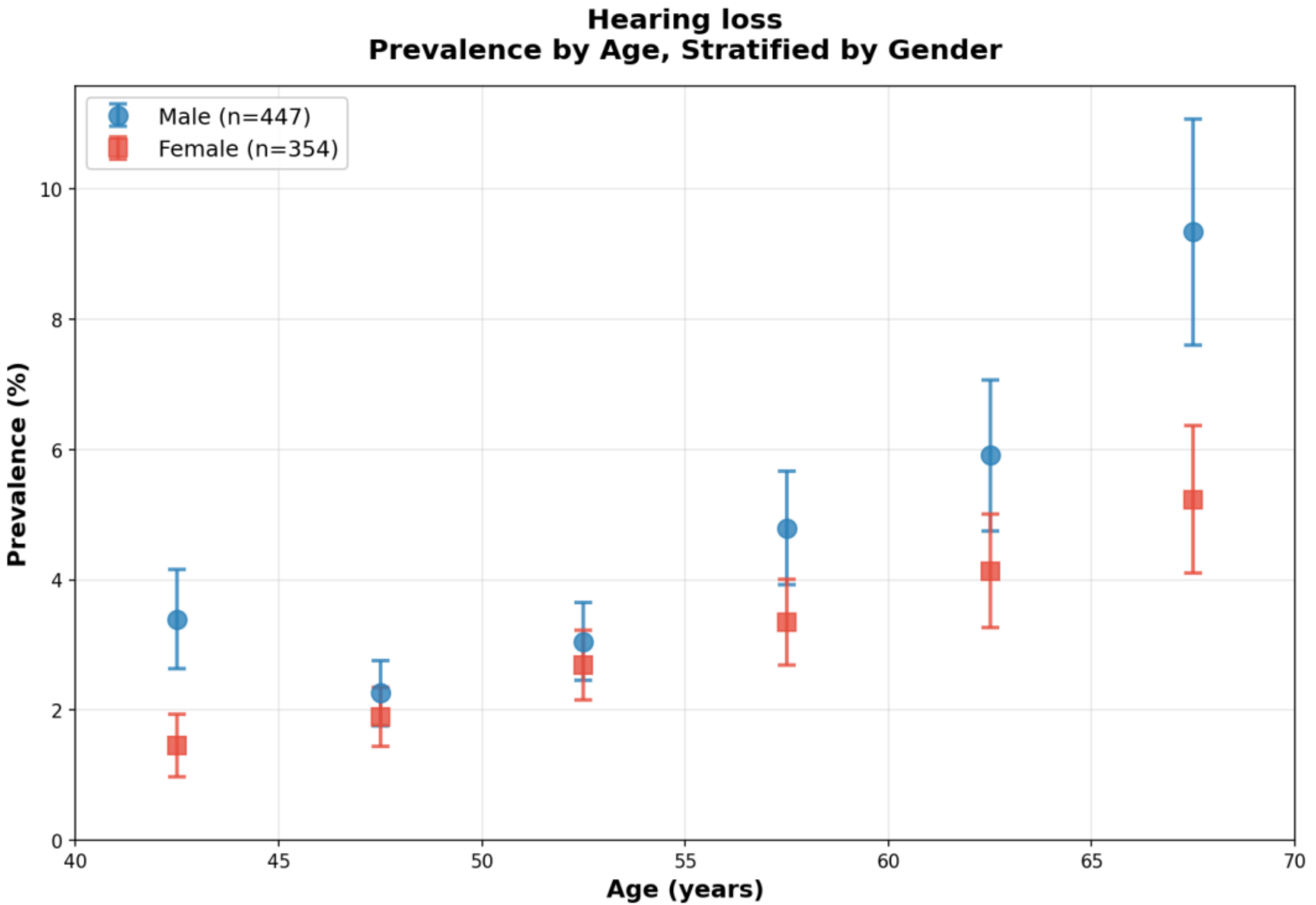

Migraine
Prevalence by Age, Stratified by Gender
Male (n=285)
Female (n=920)
Prevalence (%)
Age (years)
0
2
4
6
8
10
40
45
50
55
60
65
70

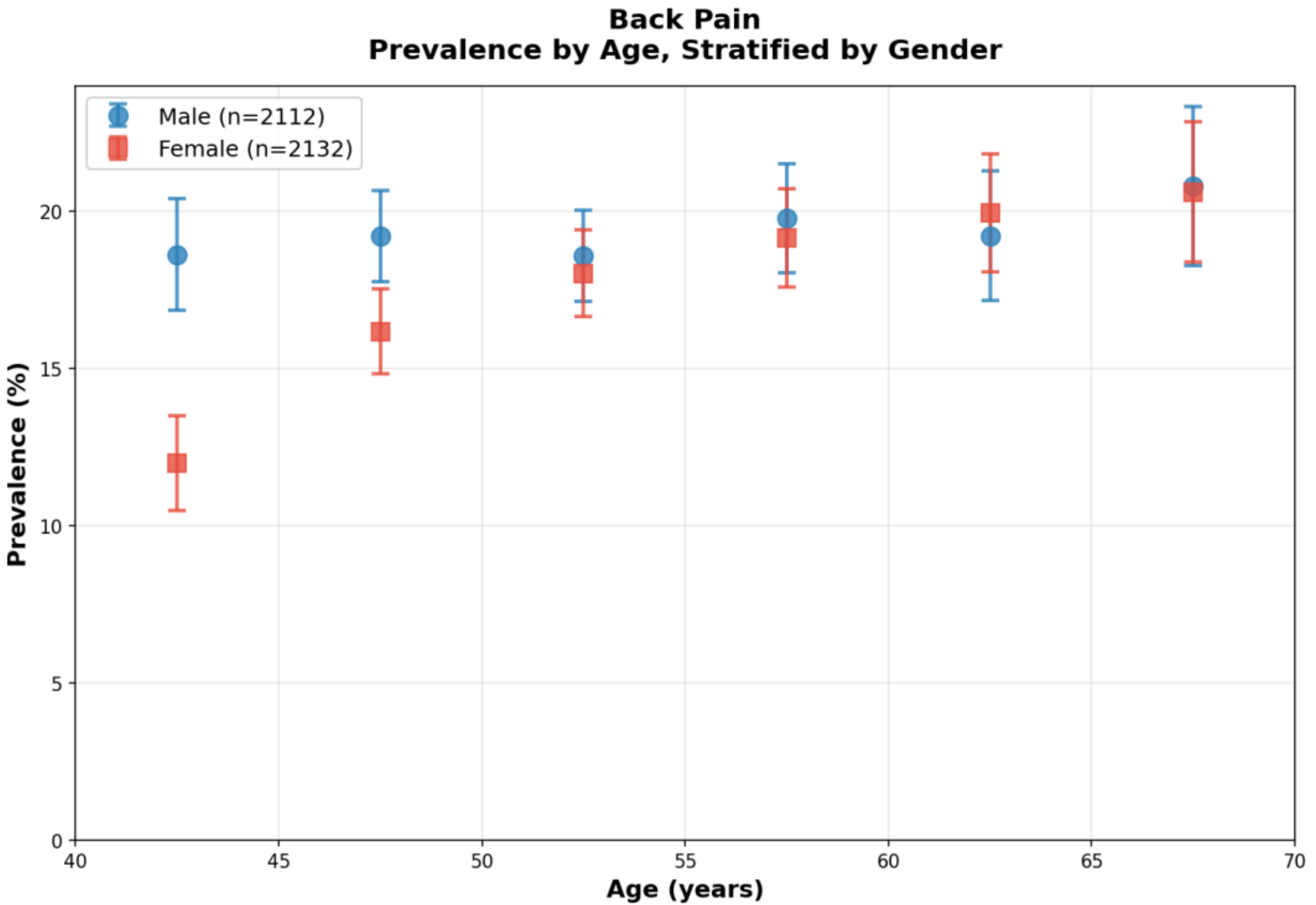

Gout
Prevalence by Age, Stratified by Gender
Male (n=193)
Female (n=15)
Prevalence (%)
Age (years)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
40
45
50
55
60
65
70

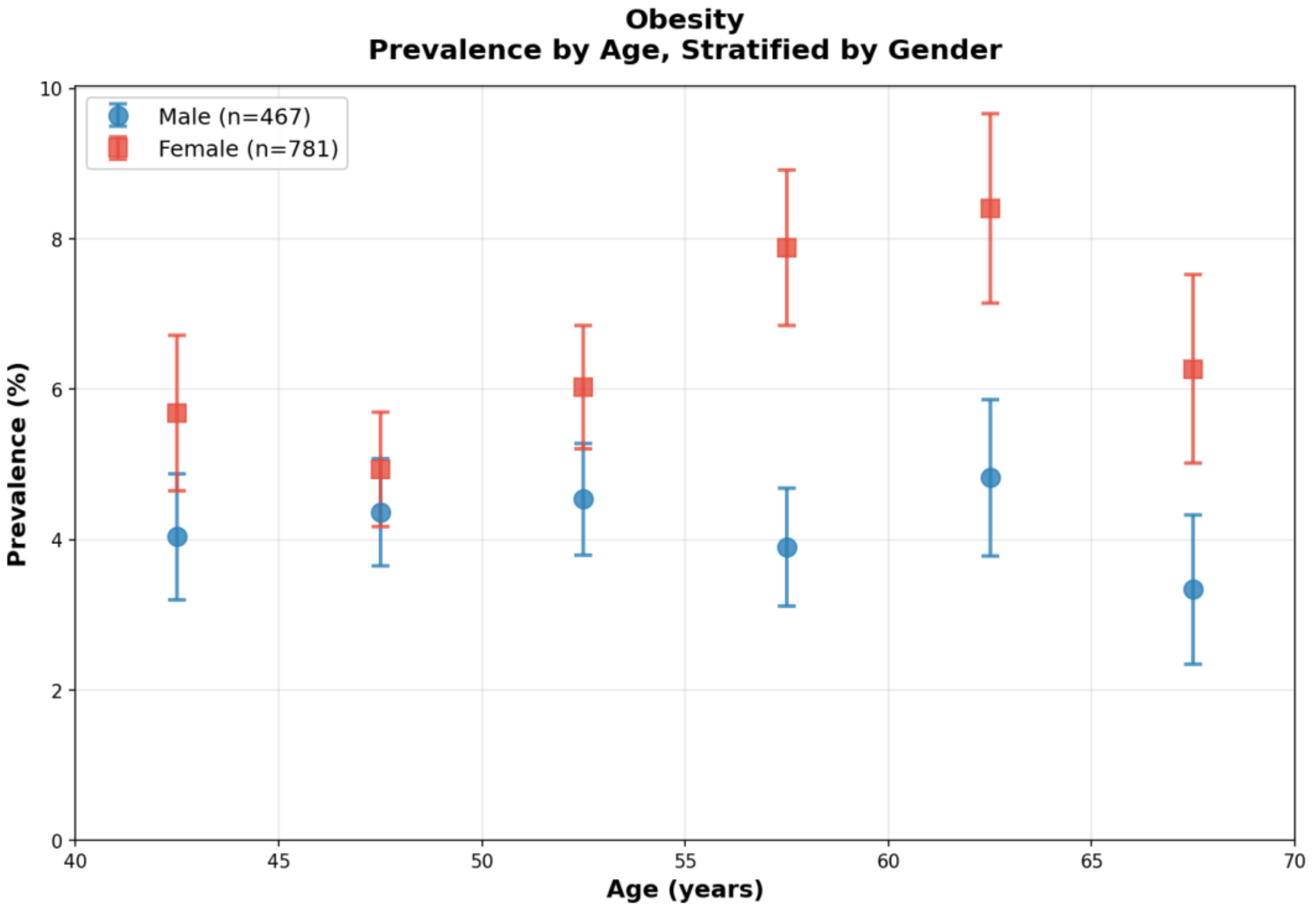

Anemia
Prevalence by Age, Stratified by Gender
Male (n=215)
Female (n=825)
Prevalence (%)
Age (years)
0
2
4
6
8
10
40
45
50
55
60
65
70

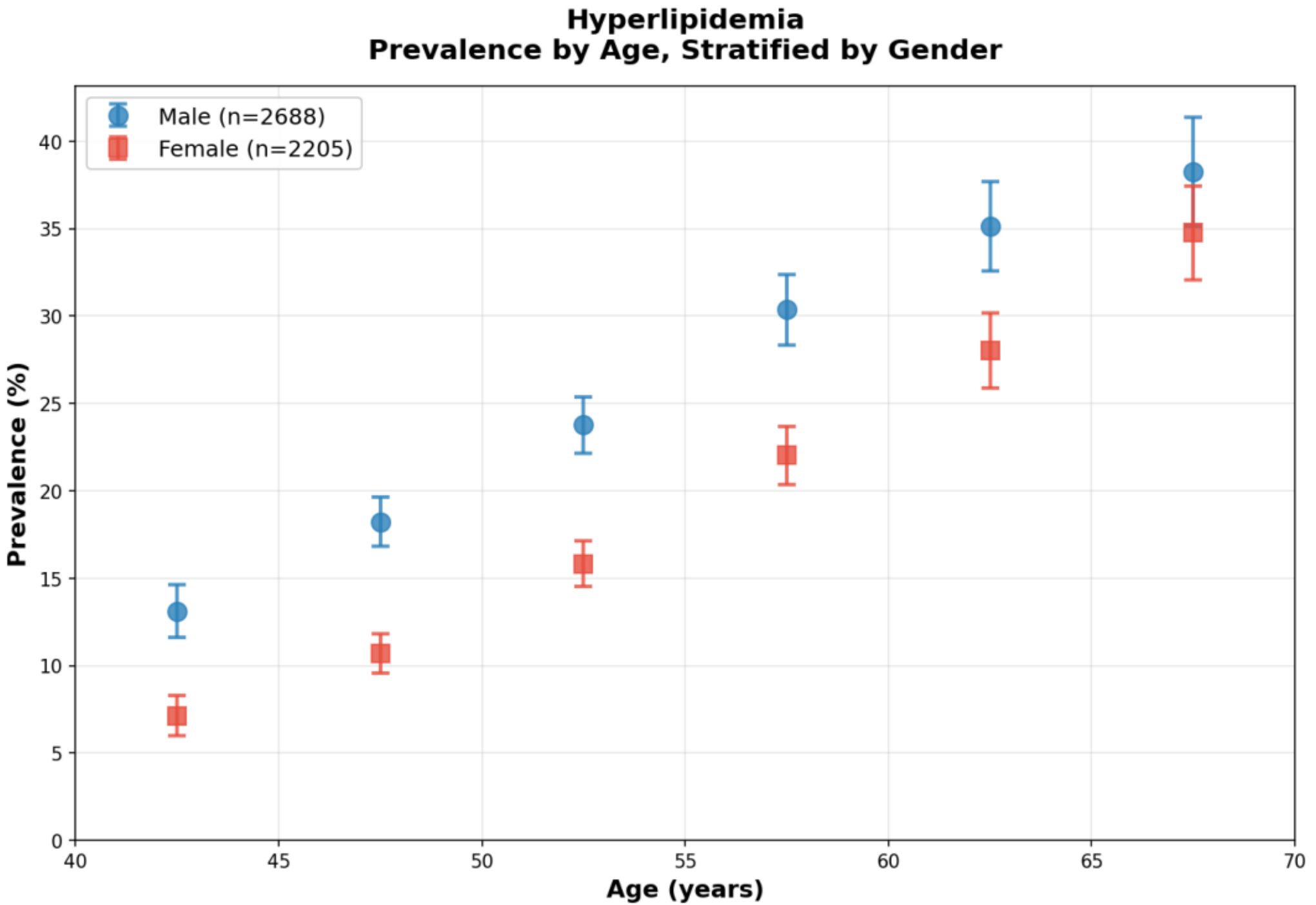

Sleep Apnea
Prevalence by Age, Stratified by Gender
Male (n=484)
Female (n=118)
Prevalence (%)
Age (years)
0
1
2
3
4
5
6
7
8
40
45
50
55
60
65
70

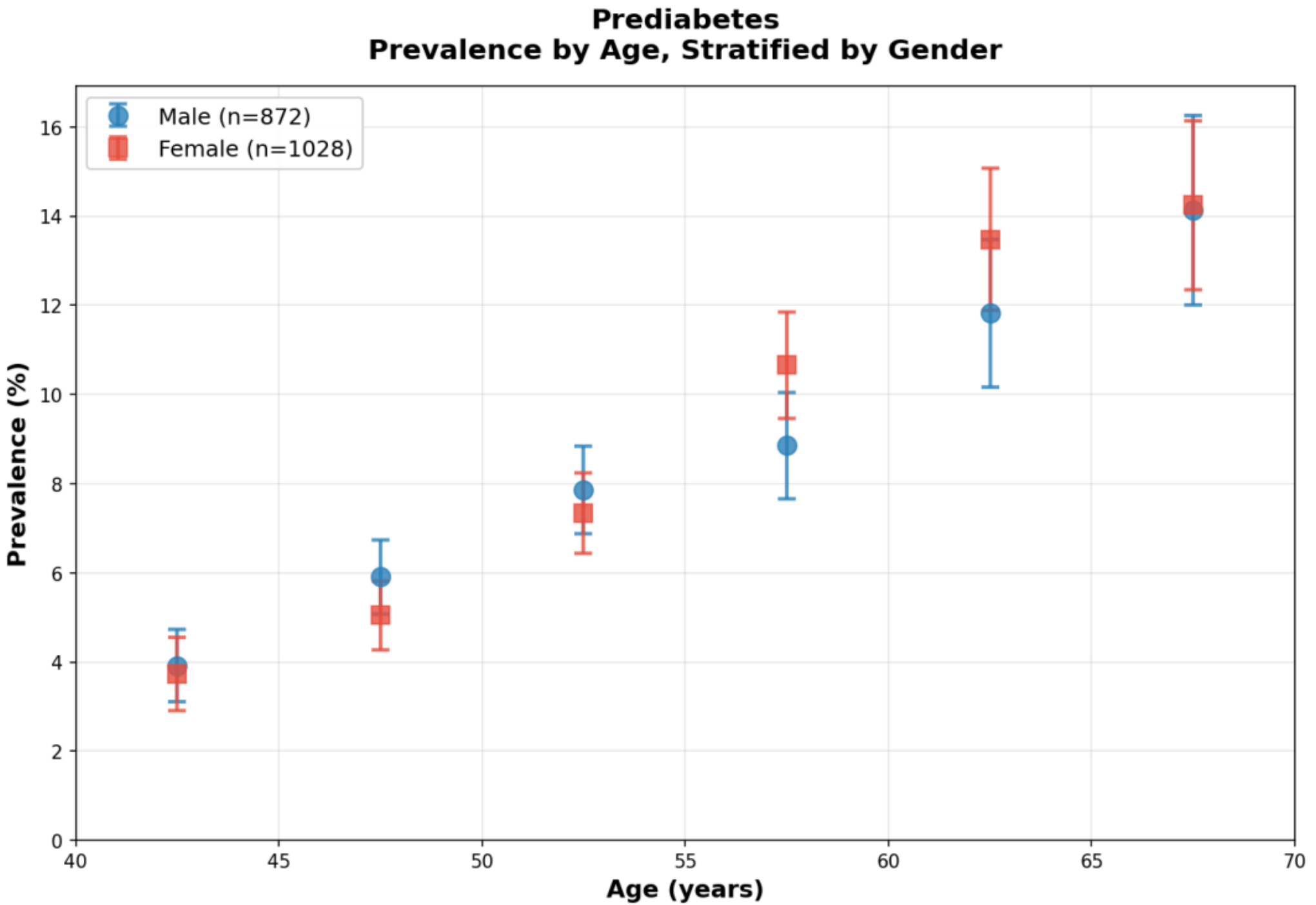

Glaucoma
Prevalence by Age, Stratified by Gender
Male (n=72)
Female (n=77)
Prevalence (%)
0.0
0.5
1.0
1.5
2.0
40
45
50
55
60
65
70
Age (years)

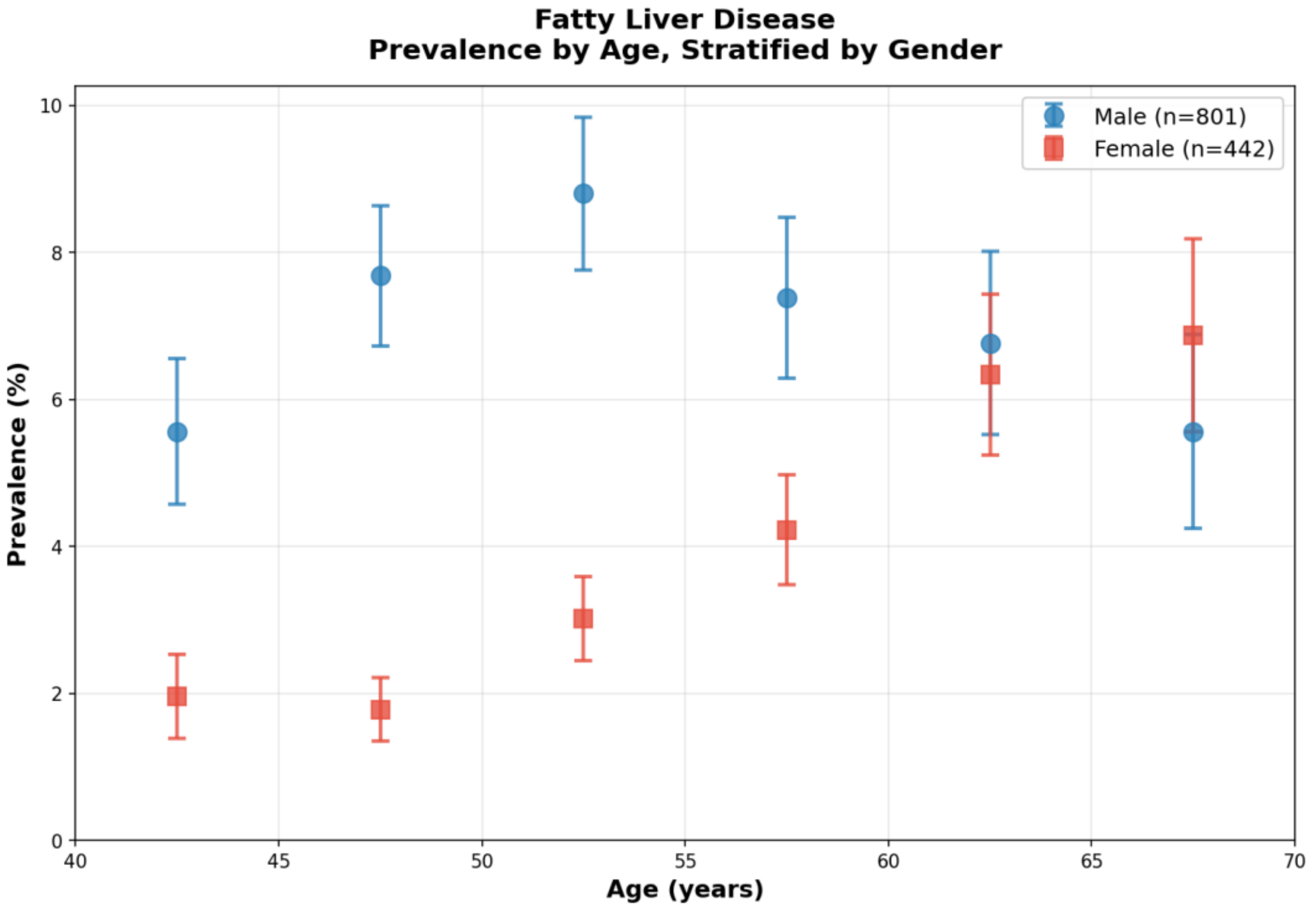

Hearing loss
Prevalence by Age, Stratified by Gender
Male (n=447)
Female (n=354)
Prevalence (%)
Age (years)
0
2
4
6
8
10
40
45
50
55
60
65
70

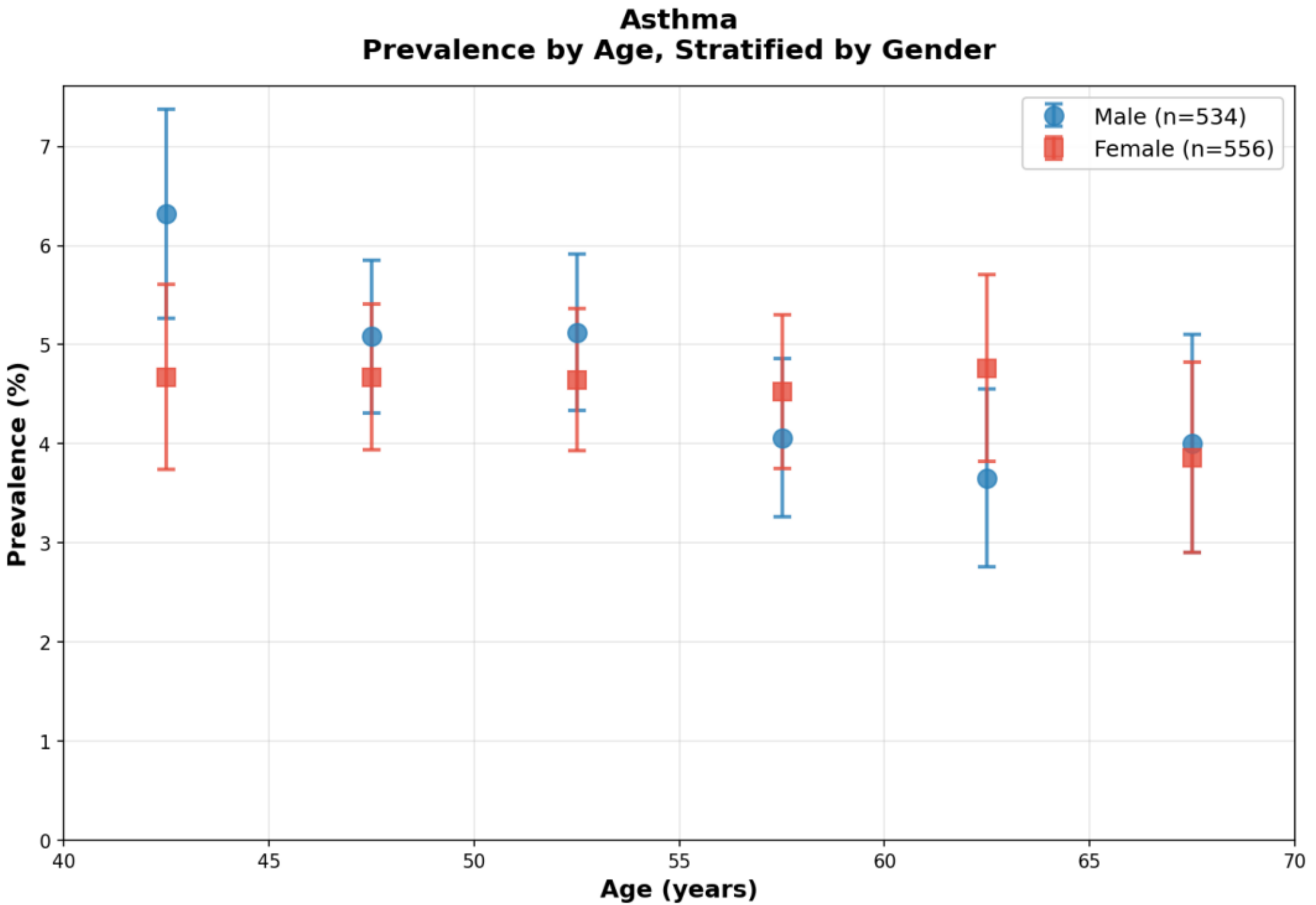

Back Pain
Prevalence by Age, Stratified by Gender
Male (n=2112)
Female (n=2132)
Prevalence (%)
0
5
10
15
20
40
45
50
55
60
65
70
Age (years)

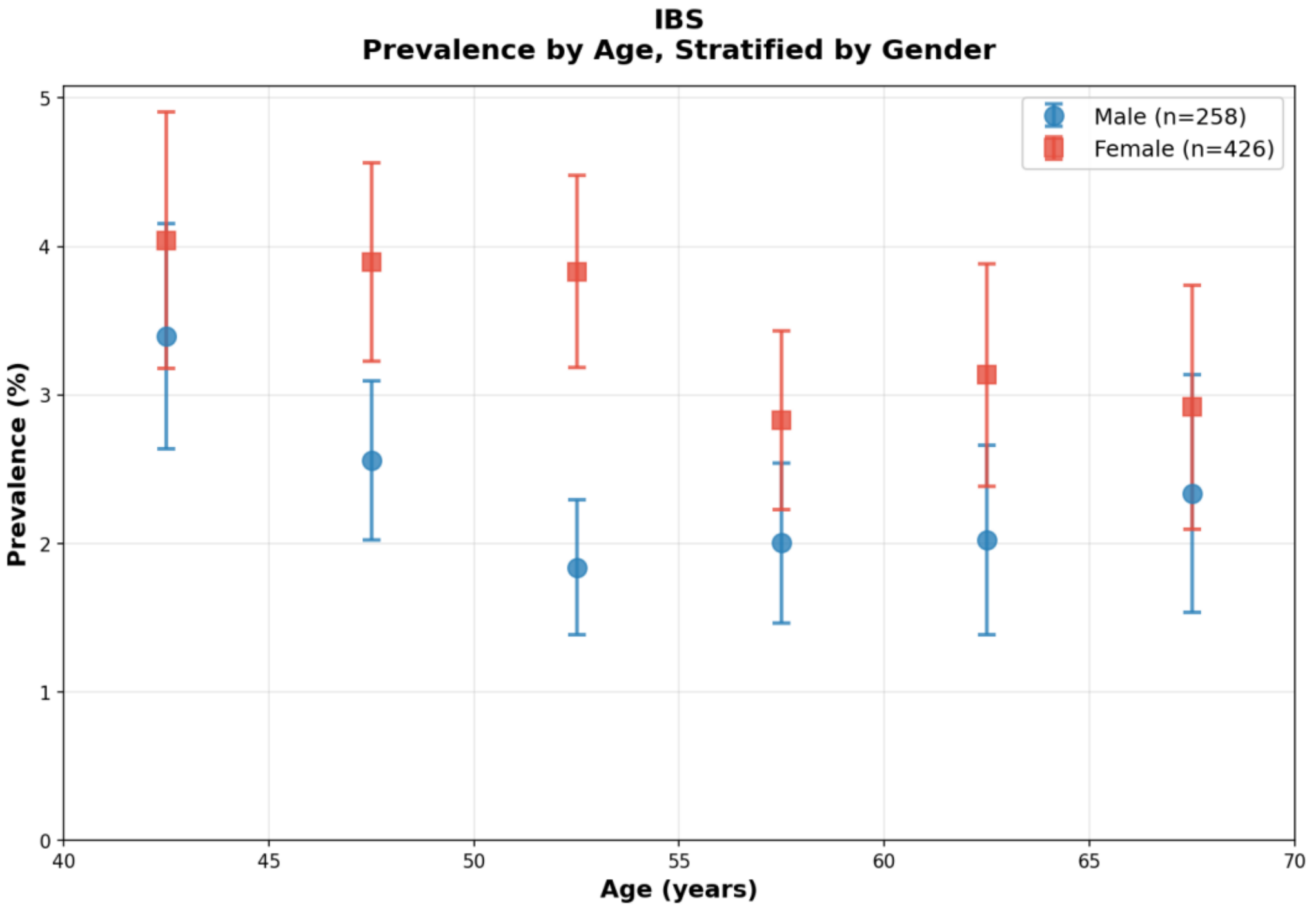

Obesity
Prevalence by Age, Stratified by Gender
Male (n=467)
Female (n=781)
Prevalence (%)
Age (years)
0
2
4
6
8
10
40
45
50
55
60
65
70

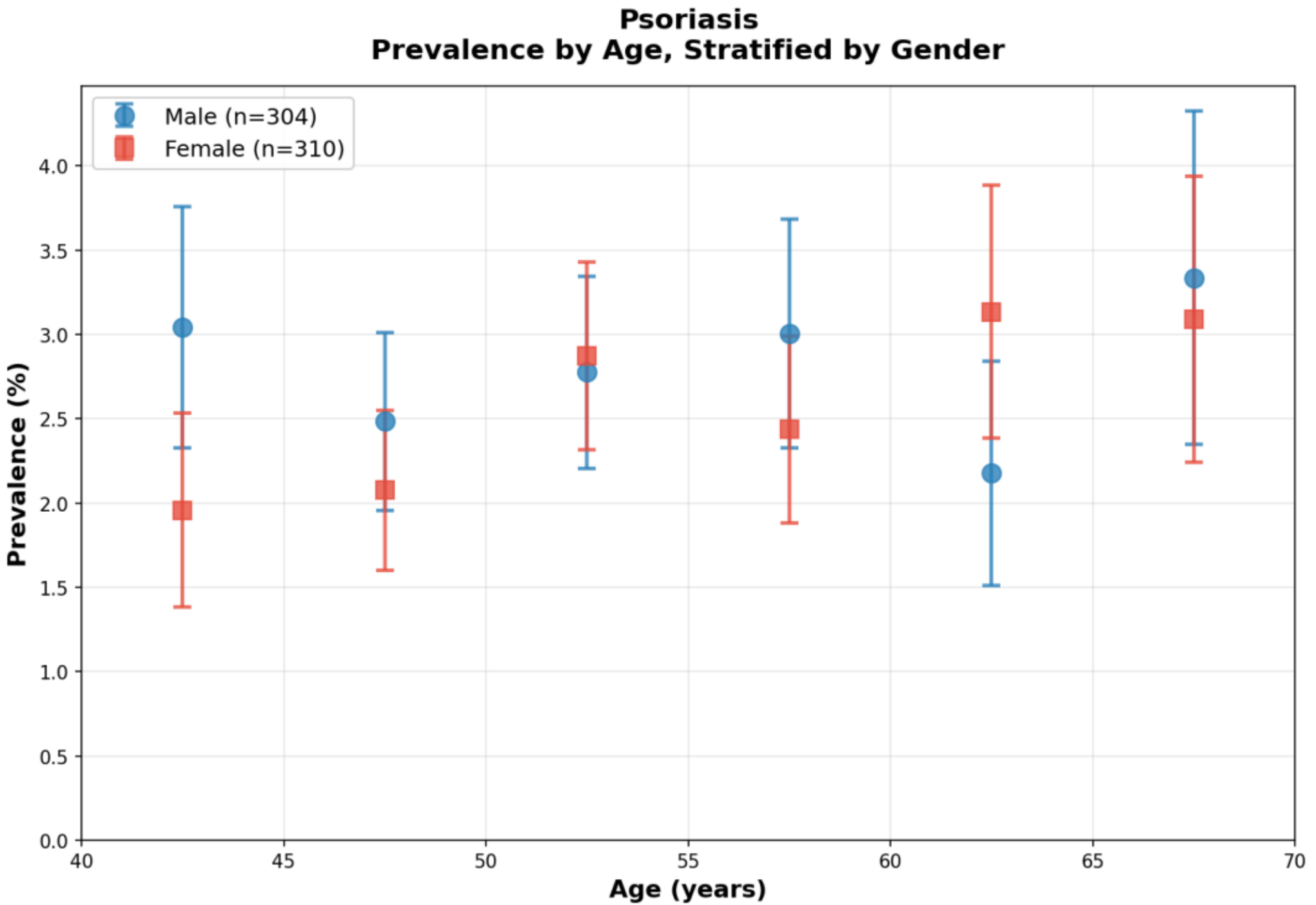

Hyperlipidemia
Prevalence by Age, Stratified by Gender
Male (n=2688)
Female (n=2205)
Prevalence (%)
Age (years)
0
5
10
15
20
25
30
35
40
40
45
50
55
60
65
70

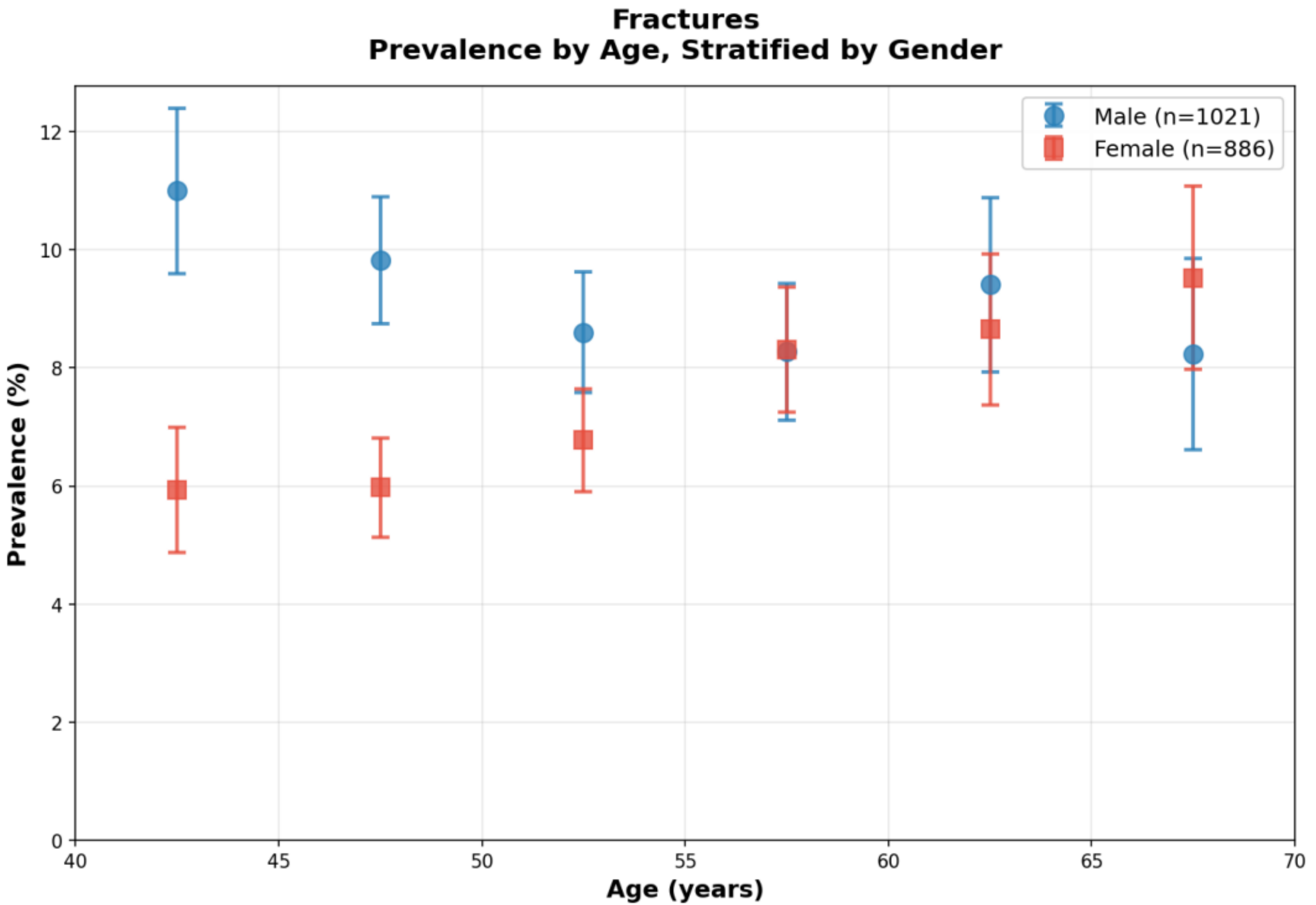

Prediabetes
Prevalence by Age, Stratified by Gender
Male (n=872)
Female (n=1028)
Prevalence (%)
Age (years)
0
2
4
6
8
10
12
14
16
40
45
50
55
60
65
70


Fatty Liver Disease
Prevalence by Age, Stratified by Gender
Male (n=801)
Female (n=442)
Prevalence (%)
Age (years)
0
2
4
6
8
10
40
45
50
55
60
65
70


Asthma
Prevalence by Age, Stratified by Gender
Male (n=534)
Female (n=556)
Prevalence (%)
Age (years)
0
1
2
3
4
5
6
7
40
45
50
55
60
65
70


IBS
Prevalence by Age, Stratified by Gender
Male (n=258)
Female (n=426)
Prevalence (%)
Age (years)
0
1
2
3
4
5
40
45
50
55
60
65
70


Psoriasis
Prevalence by Age, Stratified by Gender
Male (n=304)
Female (n=310)
Prevalence (%)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
40
45
50
55
60
65
70
Age (years)


Fractures
Prevalence by Age, Stratified by Gender
Male (n=1021)
Female (n=886)
Prevalence (%)
Age (years)
0
2
4
6
8
10
12
40
45
50
55
60
65
70

METHODOLOGY
{'='*55}

DATA PRESENTATION
• RAW DATA ONLY - No statistical models or fitted lines
• Each point = actual prevalence in that age bin
• Age bins: 5-year intervals (40-45, 45-50, etc.)
• Error bars = 95% confidence intervals (Wilson score)

PLOT INTERPRETATION
• Blue circles = Male prevalence at each age bin
• Red squares = Female prevalence at each age bin
• Y-axis = Prevalence (%) - percentage with condition
• X-axis = Age midpoint of each 5-year bin

CONFIDENCE INTERVALS
• Wilson score interval for binomial proportions
• More accurate than normal approximation for small samples
• Asymmetric intervals that respect 0-100% bounds

INCLUSION CRITERIA
• Age: 40-70 years
• Minimum 50 cases per gender for analysis
• Total n ≥ 100 for condition

ADVANTAGES OF THIS APPROACH
• No model assumptions
• Shows actual data patterns
• Easy to interpret
• Confidence intervals show uncertainty

LIMITATIONS
1. No statistical testing of trends
2. Cannot extrapolate beyond observed ages
3. Some conditions are sex-specific
4. Cross-sectional design (not longitudinal)
5. Survival bias may affect older ages

SAMPLE SIZES
• Total: 23,473 subjects
• Female: 12,313 (52%)
• Male: 11,152 (48%)

KEY FINDINGS BY CONDITION CATEGORY
{'='*55}

CARDIOVASCULAR
• Hypertension: Clear increase with age in both genders
  Males have higher prevalence at all ages
• Ischemic Heart Disease: Increases with age, much higher
  in males

METABOLIC
• Hyperlipidemia: Dramatic increase with age in both
  Males start higher but females catch up
• Diabetes: Increases with age in both genders
• Prediabetes: Similar pattern to diabetes

MENTAL HEALTH
• Depression: DECREASES with age in both genders
  Likely cohort effect (younger people diagnosed more)
• Anxiety: Similar decreasing pattern
• ADHD: Strong cohort effect visible

MUSCULOSKELETAL
• Osteoarthritis: Increases with age, steeper in females
• Back Pain: High prevalence, increases slightly with age
• Gout: Much more common in males, increases with age

AUTOIMMUNE/INFLAMMATORY
• Allergy: DECREASES with age in both genders
  Another cohort effect
• Psoriasis: Relatively stable across ages

SENSORY
• Hearing loss: Increases with age, higher in males
• Glaucoma: Increases with age in both genders

NOTES ON INTERPRETATION
• These are RAW DATA - no statistical modeling
• Patterns visible to the eye are real data patterns
• Gender differences may reflect:
  - Biological differences
  - Healthcare-seeking behavior
  - Diagnostic patterns
  - Survival differences
• Always check sample sizes (n) in legend